SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Delhi-NCR Campus, Modinagar, Ghaziabad (U.P.)

ASSIGNMENT- IV

(B. Tech. III -Year, VI-SEMESTER)

SUBJECT NAME: DATA SCIENCE

SUBJECT CODE: 21CSS303T

Submitted To:

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Section- A

(2025 - 2026)

) Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

andling large datasets involves efficient techniques and ogramming practices manages performance and resources.

roblem with Large data:

data quality issues

Security & privacy concerns

High cost

Data integrity issues

Difficulty in analysis

chniques for handling data

Choose the right algorithm

→ Efficient algorithms improve performance

→ Examples: online algorithms, map reduce

Choose the right data structure

→ Use appropriate structure like:-

- Trees
- Hash tables
- Sparse data structure

Importance :

- Improves data quality and accuracy
- Makes data ready for analysis
- Helps in better decision-making
- Saves time in analysis
- Handles missing and inconsistent data.

12) Explain string manipulation in python with suitable code examples.

String manipulation refers to preprocessing and modifying text data using python functions and methods.

) Basic string methods

```
val = 'a, b, c'
# split string
val.split(',')
# ['a', 'b', 'c']
# Removes spaces
pieces = [x.strip() for x in val.split(',')]
# join strings
':'.join(pieces)
```

) Searching operations

```
val = 'a, b, c'
'c' in val   # True
```

```
al.find(',') #1
l. count(',') # 2
```

ii) Replace operation

```
val. replace(',', '::')
val. replace(',', '')
```

Regular Expressions

```
import re
text = "foo bar\t baz"
re. split('\st', text)
# ['foo', 'bar', 'baz']
```

Pandas vectorized string functions

```
import pandas as pd
import numpy as np
data = pd. series({
        'Dave': 'dave@google.com'
        'Steve': 'steve@google.com'
        'Rob': 'rob@google.com'
        'Wes': 'wes@google.com' })
data. str. contains('google')
```

1) Explain the concept of data wrangling and its importance in data analysis.

Data wrangling is the process of gathering, collecting, cleaning and transforming raw data into a useable format for analysis. It is also called Data Munging.

Steps of Data Wrangling:

i) Data acquisition - collect data from different sources

ii) Data cleaning - Remove errors, missing values

iii) Data exploration - Understand structure and content

iv) Data transformation - Convert into suitable format

v) Data Loading - Load into analysis tools

Key operations:

- clean
- Transform
- Merge
- Reshape

) Choose the right tools

-> Use optimized tools such as

- Numpy
- Numba
- Blaze
- Cython

neral Programming Tips

Do not reinvent the wheel

Get maximum use of hardware

Reduce computational needs

) Explain different data transformation techniques used in preprocessing.

ta transformation is the process of converting data into suitable format for analysis.

chniques :

removing duplicates - df.drop_duplicates()

ransforming data (mapping) - df['col'] = df['col'].map(lambda x: x*2)

eplacing values - df.replace(-999, 0)

Renaming Index/Columns - df.rename(columns={'old': 'new'})

Binning (Discretization) - pd.cut(data, bins=[0, 50, 100]

Outlier detection - using IQR or Z-score

) Creating dummy variables - pd.get_dummies(df['col'])